PyGame Flappy Bird Tutorial - Part 7

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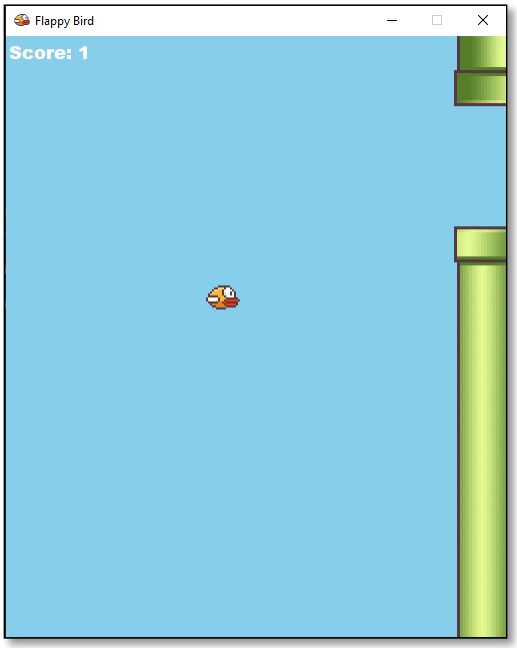
Time required: 30 minutes

Preview of the Game

Here's a sneak peak of the game that we are going to work on.

Flappy Bird Demo Video



Full Game

Time to finish up the game with scoring and sounds.

1. Save **flappy_bird_6.py** as **flappy_bird_7.py**

2. Modify the existing code.
3. Add sleep to pause the game for a moment.

```python
"""
    Name: flappy_bird_7.py
    Author:
    Date:
    Purpose: Flappy Bird Clone in OOP
"""
# pip install pygame-ce
# Import pygame library
import pygame
# pip install pygame-menu
import pygame_menu as pm
# Import exit for a clean program shutdown
from sys import exit
from random import randint
from time import sleep
import config
```

4. Initialize the pygame mixer to optimize the sounds.

```python
class FlappyBird:
    def __init__(self):
        # pre initalize mixer with larger buffer size for better performance
        pygame.mixer.pre_init(
            44100,  # frequency (Hz)
            16,     # bit depth
            2,      # number of channels, 1 mono, 2 stereo
            4096    # buffer size, larger to optimize music playback.
        )

        # Initialize pygame engine
        pygame.init()
```

5. Load the background music, set the volume and start the music playing. Setup a few variables.

```python
        self.init_bird()
        self.init_pipes()

        # Load flappy bird program png icon
        self.bird_ico = pygame.image.load(
            "./assets/flappy_bird_ico.png").convert_alpha()

        pygame.display.set_icon(self.bird_ico)

        # Load background music file into memory
        pygame.mixer.music.load('./assets/flying-minimal.mp3')

        # Set volume to 30%, range from 0.0 (mute) to 1.0 (full volume)
        pygame.mixer.music.set_volume(0.3)

        # Play in a loop until stopped
        pygame.mixer.music.play(-1)

        self.score = 0
        self.game_over = False
        self.pass_pipe = False
        self.score_font = pygame.font.SysFont("arialblack", 18)
```

6. Add score counted variable to the end of the init pipes method.

```python
        # Set lower pipe vertical location
        self.pipe_lower_rect.top = self.pipe_upper_rect.bottom \
            + self.pipe_gap_size

        # Set score counted to false
        self.score_counted = False
```

7. Add the same variable to the end of the reset pipes method.

```python
        # Set initial X off screen to right
        self.pipe_upper_rect.left = config.WIDTH
        self.pipe_lower_rect.left = config.WIDTH

        # New set of pipes, reset score counter
        self.score_counted = False
```

8. Add the display score method.

```python
# ------------------- DISPLAY SCORE ---------------------------------------#
    def display_score(self):
        """Display the score on the screen"""
        # Create text image for score display
        text = self.score_font.render(f"Score: {self.score}", True, "white")
        self.surface.blit(
            text,   # Image to display
            [3, 3]  # x , y to display the image
        )
```

9. Make some modifications to the display game over menu.

```python
# ------------------------- DISPLAY GAME OVER ----------------------------#
    def display_game_over(self):
        """Display game over menu using the Pygame Menu library"""
        # Stop background sound
        pygame.mixer.music.stop()

        # Play crash sound
        crash = pygame.mixer.Sound('./assets/crash_short.mp3')
        crash.play()
        crash.set_volume(0.3)

        # Wait 2 second while crash plays
        sleep(2)

        # Define a menu object for the game over screen
        game_over = pm.Menu(
            title="Game over",      # Set title menu to "Game over"
            width=config.WIDTH,     # Set to width of game surface
            height=config.HEIGHT,   # Set to height of game surface
            # Set the theme of the menu to an orange color scheme
            theme=pm.themes.THEME_ORANGE
        )

        # Display final score
        game_over.add.label(f"Score: {self.score}")

        # Add label to provide space between buttons
        game_over.add.label("")

        # Add a button to the game over menu for exiting the game
        game_over.add.button(
            title="Play Again?",    # Button text
            action=main             # Call main() to start over
        )

        # Add label to provide space between buttons
        game_over.add.label("")

        # Add a button to the game over menu for exiting the game
        game_over.add.button(
            title="Exit",           # Button text
            action=pm.events.EXIT   # Exit the game when clicked
        )

        # Run the main loop of the game over menu on the specified surface
        game_over.mainloop(self.surface)
```

10. Some game_loop modifications.

```python
# --------------------------- GAME LOOP ----------------------------------#
    def game_loop(self):
        """Infinite game loop"""
        while not self.game_over:
            self.check_events()
            self.detect_collision()
            # Simulate gravity by moving the bird down
            # unless the UP key is pressed
            # Reset gravity to 3 each time through the loop
            gravity = 3

            # Get list of keys being pressed
            key_input = pygame.key.get_pressed()

            # If up cursor pressed, move up 5 pixels
            if key_input[pygame.K_UP]:
                gravity -= 5

            # ------------------- INCREASE DIFFICULTY ---------------------#
            # Adding difficulty relative to score
            # Increase the speed and decrease the gap of blocks
            if 5 <= self.score < 10:
                self.pipe_move = 5
                self.pipe_gap_size = self.bird_rect.height * 4

            elif 10 <= self.score < 20:
                self.pipe_move = 7
                self.pipe_gap_size = self.bird_rect.height * 3.5

            # ---------------------- SCORING ------------------------------#
            # If the bird makes it past the pipes, increase score
            if self.bird_rect.left > self.pipe_upper_rect.right \
                    and not self.score_counted:

                # Increase score
                self.score += 1

                # Track whether the current set of pipes have had a score
                self.score_counted = True
```

Example run:

A complete game!

There is always room for improvement.

1. Add more difficulty levels.
2. Keep track of the highest score.
3. Add music.
4. Etc.

Assignment Submission

Attach all program files to the assignment in BlackBoard.

Assignment Submission

1. Attach all tutorials and assignments.
2. Attach screenshots showing the successful operation of each tutorial program.

3. Submit in Blackboard.